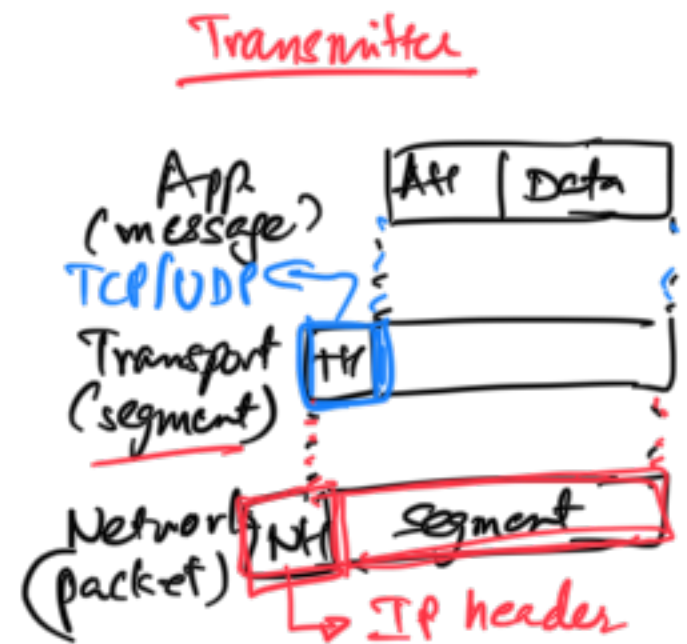


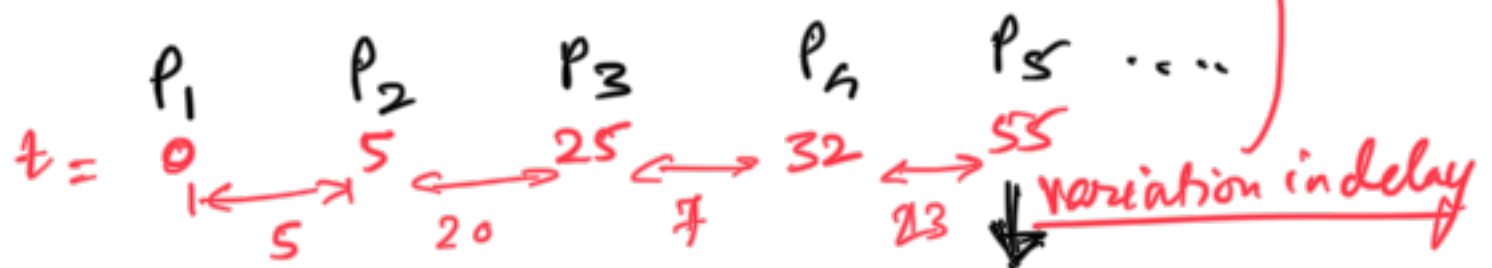
* Functions of Network layer

- source to dest packet delivery
 - Src IP
 - dest IP
- Routing
 - ↳ select best path to the dest
- Fragmentation
- :



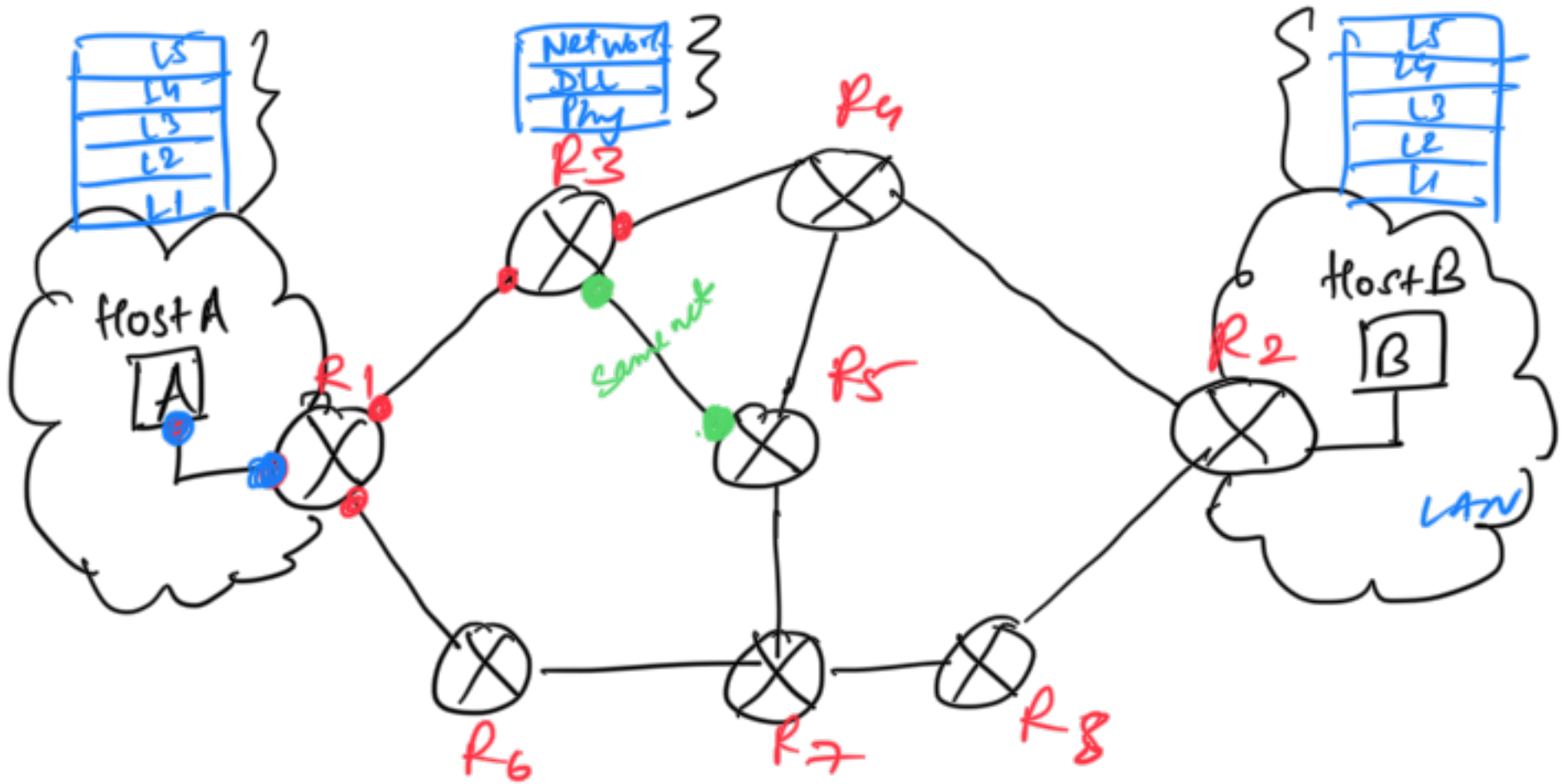
* Services that can be requested by Transport layer from the network layer

- reliable delivery from src to dest
- guaranteed bandwidth / data rates
- bounded end-to-end delay
- bounded limits on jitter



* Example network

Q. How many layers of the protocol stack does each node work on?



Routing table structure

Pkt

NH

Dst IP addr

Net addr	Next hop	Metric
175.65.0.0/16		

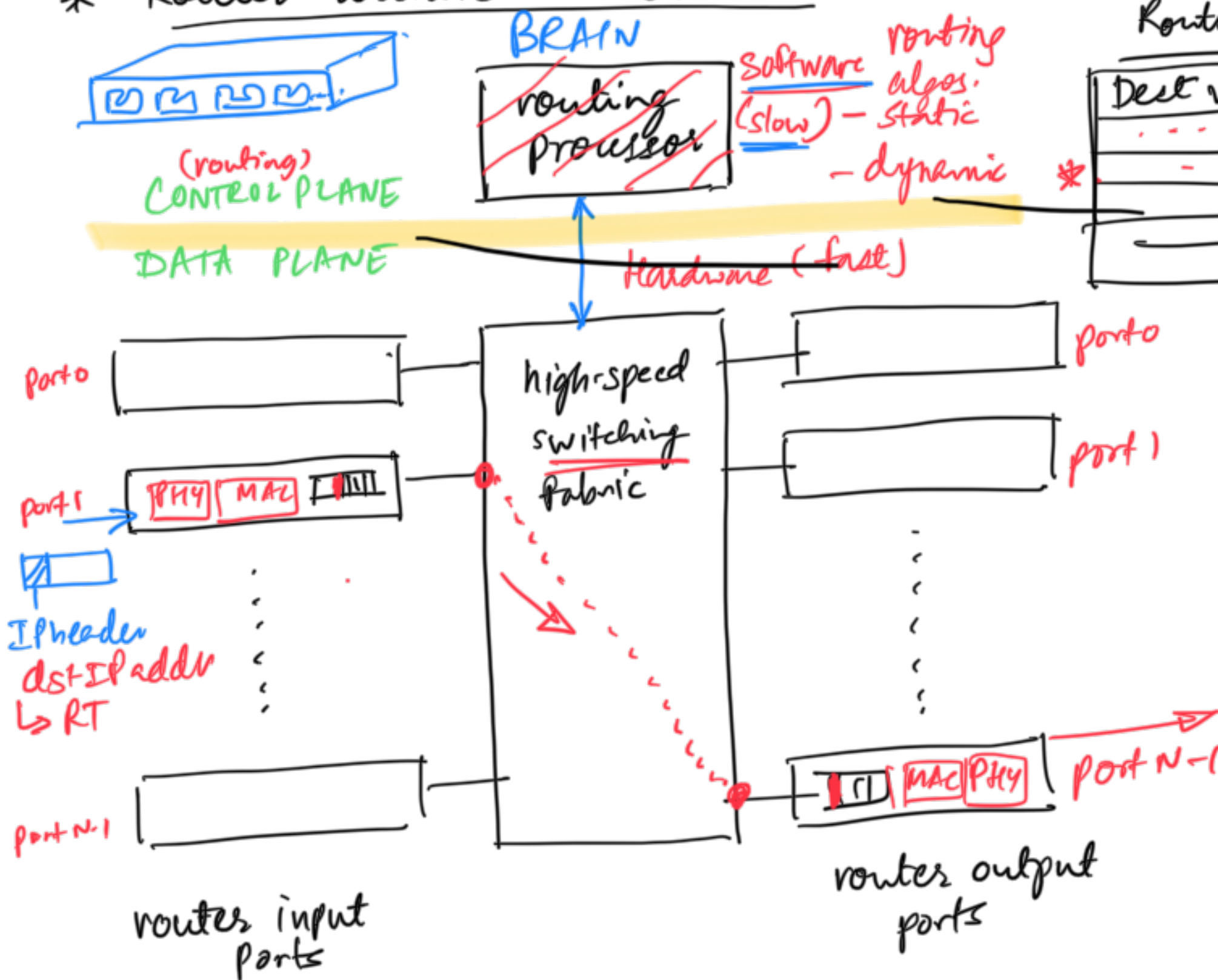
Classless Interdomain routing

CI DR notation

k: # of leading 1's in the subnet mask (# of netid bits)

Q. Who gets an IP address?

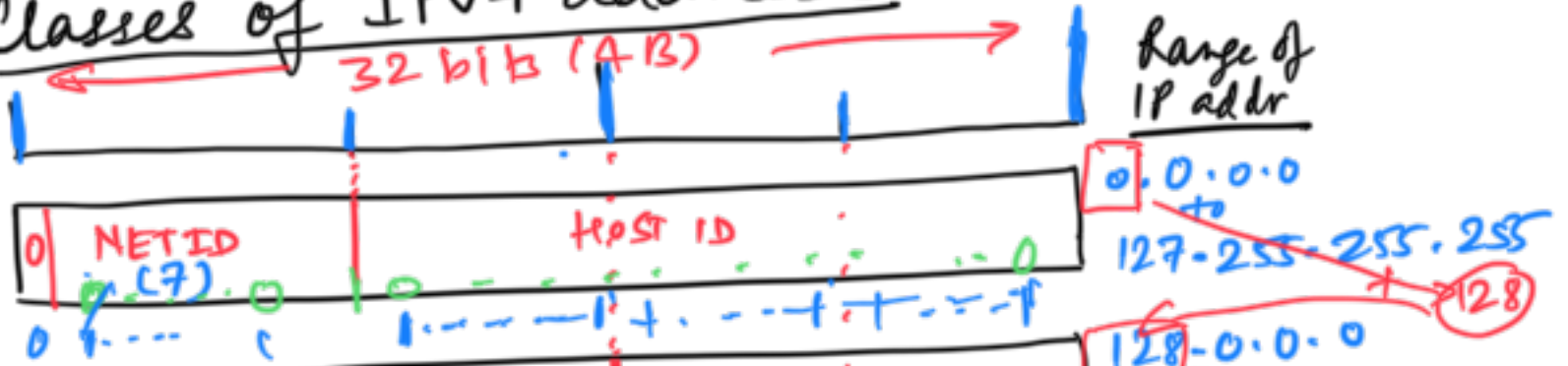
* Router architecture overview



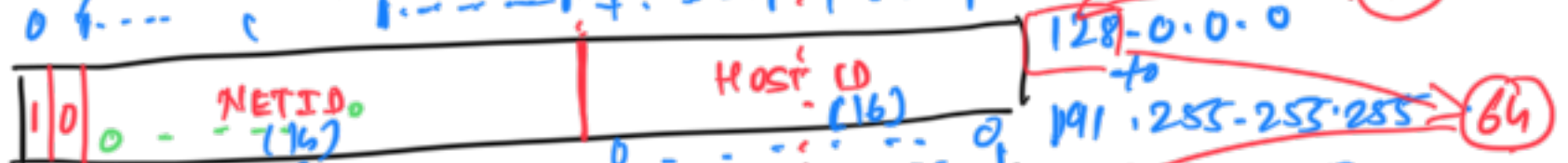
* IP addressing IPv4 : 32 bits (4B) IPv6 : 128 bits
 • dotted decimal notation
 D.D.D.D

* Classes of IPv4 addresses

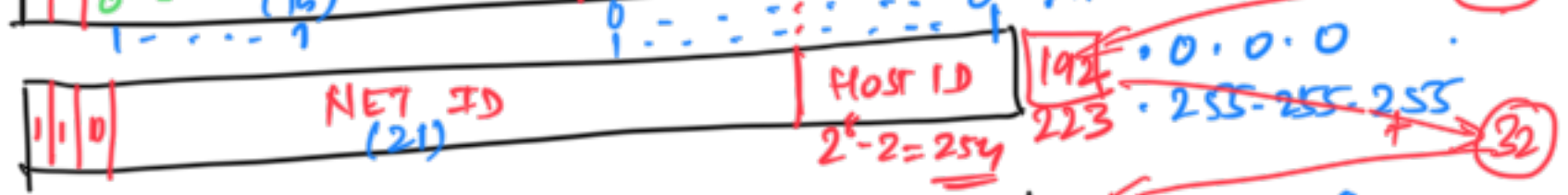
max networks
Class A 2^7



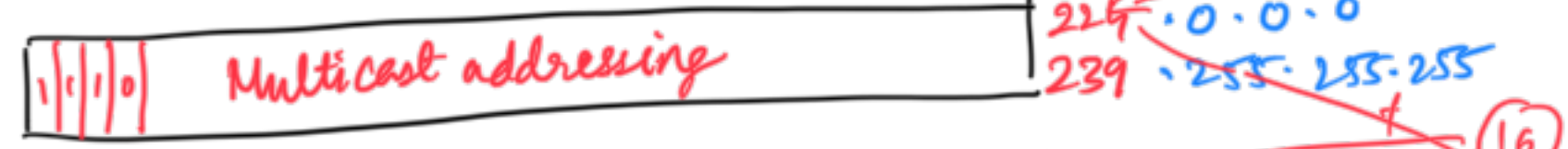
Class B 2^{14}



Class C 2^{21}



Class D



Class E



2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	
128	64	32	16	8	4	2	1	
0	0	0	0	0	0	0	0	$\Rightarrow 0$
1	1	1	1	1	1	1	1	$\Rightarrow 255 (2^8 - 1)$
0	1	1	1	1	1	1	1	$\Rightarrow 255 - 128 = 127$
0	0	0	0	0	0	0	0	$\Rightarrow 128$
1	0	1	1	1	1	1	1	$\Rightarrow 191$

Example

Boolean AND
^

Netid Hostid
175. 65. | 20. 2
1-----1 | 0-----0
255. 255. | 0. 0

... Class B IP addr
^

Subnet mask

↓

175. 65. | 0. 0

Network addr

175. 65. | 255. 255

Broadcast addr

AND



Broadcast addr
11111111
eg MAC addr broadcast
ff:ff:ff:ff:ff:ff
(48 bits)

	Netid	Hostid
Netaddr	(As it is)	(All 0's)
Subnet mask	All 1's	All 0's
Broadcast address for a network	As it is	All 1's

- * Maximum no. of networks = 2^k k: # of netid bits
- Class A: k=7
Class B: k=14
Class C: k=21
- * Maximum no. of hosts on a network = $2^n - 2$ n: # of hostid bits
- minus 2: 0...0 net addr
 1...1 broadcast addr

	Net id	Host id
Network addr		
Broadcast addr		
Subnet mask		

Q. What is the structure of routing table? why?

